

Control PCB

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Project Objective

This document details the custom printed circuit board (PCB) designed to control an automated stage-top bioreactor. The system is engineered to provide precise environmental regulation and mechatronic actuation required for mammalian cell culture, tissue handling, and 3D organoid generation.

1 Hardware Status

The core PCB design is complete, and all design rule checks (DRC) have been successfully verified. All electrical errors have been resolved, with high-current switching nodes physically isolated from precision analog sensing zones. No further schematic modifications will be made to this baseline control board. Future hardware revisions will be strictly limited to the integration of external CO_2 and N_2 gas regulation manifolds, should those modules require direct PCB control rather than standalone drivers.

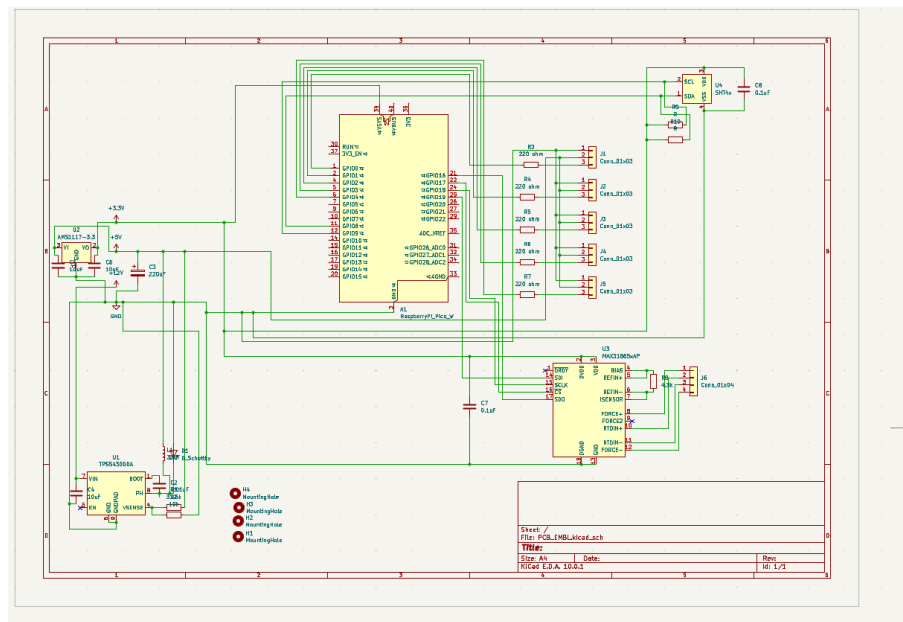


Figure 1: Finalized system schematic showing completely resolved electrical connections.

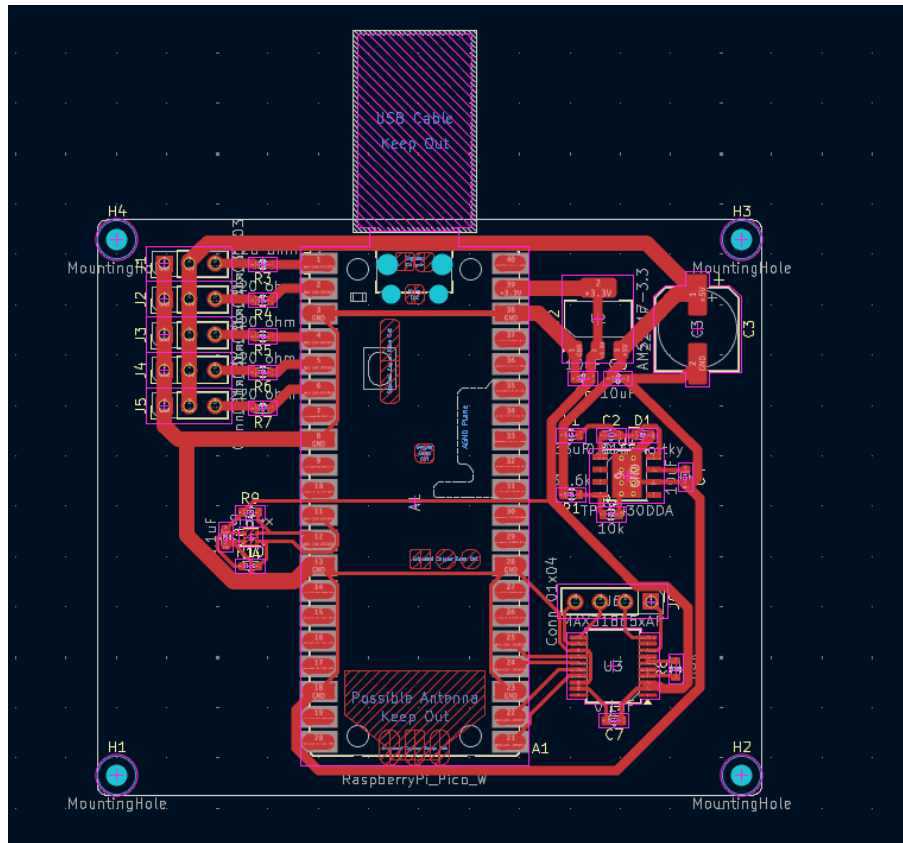


Figure 2: Finalized PCB layout with routed power and signal traces, prior to ground plane pour.

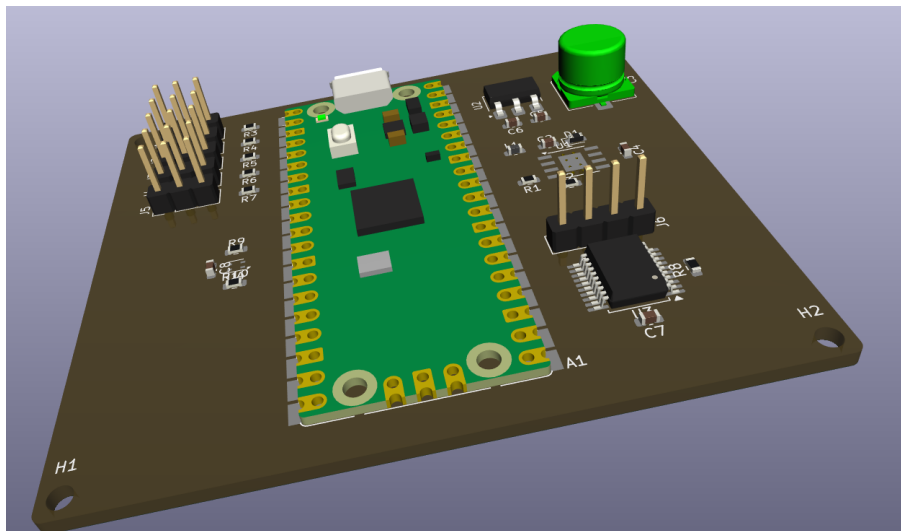


Figure 3: 3D mechanical render of the assembled stage-top bioreactor control PCB.

2 Microcontroller



Figure 4: Raspberry Pi Pico W microcontroller module.

The central processing unit is the **Raspberry Pi Pico W** (A1). It handles all PWM generation for the servo motors, I^2C /SPI sensor communication, and wireless telemetry for bioreactor monitoring.

3 Sensor Specifications

3.1 Temperature Regulation (MAX31865 & PT1000)



Figure 5: MAX31865 RTD-to-Digital Converter.



Figure 6: PT1000 Platinum Resistance Thermometer probe.

High-precision thermal measurement for the 37.0°C biological target is handled by the **MAX31865** RTD-to-Digital Converter (U3).

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- **Probe:** Interface designed for a Class A **PT1000** Platinum Resistance Temperature Detector via a 4-pin screw terminal (J6). The PT1000 has a nominal resistance of $1000\ \Omega$ at 0°C and provides highly linear resistance changes proportional to temperature.
- **Topology:** 4-wire ratiometric configuration (FORCE+, FORCE-, RTDIN+, RTDIN-) to completely eliminate lead-wire resistance errors.
- **Reference Resistor:** $4.3\ \text{k}\Omega$ (0.1% tolerance) precision resistor (R8) bridging the BIAS and REFIN+ pins to define the ADC reference voltage.
- **Data Interface:** SPI bus connected to Pico W GPIO16 (MISO), GPIO17 (CS), GPIO18 (SCLK), and GPIO19 (MOSI).

3.2 Ambient Humidity (SHT4x)



Figure 7: SHT4x series digital temperature and humidity sensor package.

Relative humidity monitoring to maintain the 95% RH incubator environment is handled by the **SHT4x** digital sensor (U4).

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- **Range & Accuracy:** Operating range of 0 ... 100% Relative Humidity and $-40 \dots 125^{\circ}\text{C}$. Typical RH accuracy is $\pm 1.5\%$.
- **Data Interface:** I^2C bus connected to Pico W GPIO8 (SDA) and GPIO9 (SCL).
- **Pull-ups:** $10\text{ k}\Omega$ pull-up resistors (R9, R10) referencing the 3.3V logic rail.

4 Power Architecture & Capacitance Matrix

The board utilizes a dual-stage power regulation system to drop a 12V DC input to a 5V high-current rail for actuation, and a filtered 3.3V logic rail for the microcontroller and sensors.

4.1 Stage 1: 5V Buck Converter (TPS5430DDA)

Handles the primary voltage step-down and supplies up to 3A for the servo motors.

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- **C4 ($10\mu\text{F}$ SMD):** Input decoupling capacitor placed flush against VIN and GND pins to absorb 12V high-frequency transient switching spikes.
- **C2 ($0.01\mu\text{F}$ SMD):** Bootstrap capacitor bridging BOOT and PH pins to drive the internal high-side MOSFET.
- **D1 (Schottky Diode):** Catch diode to provide a current path during the switch off-time.
- **L1 ($33\mu\text{H}$ SMD):** Primary energy storage inductor.
- **C3 ($220\mu\text{F}$ Electrolytic):** Bulk output filter capacitor placed immediately after L1 to smooth switching ripples and provide instantaneous current during sudden servo actuation.

4.2 Stage 2: 3.3V Linear Regulator (AMS1117-3.3)

Provides a clean, low-noise voltage supply isolated from the motor rails.

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- **C5 ($10\mu\text{F}$ SMD):** Input filter capacitor to strip residual buck converter switching noise from the 5V line before it enters the LDO.

- **C6 (10 μF SMD):** Output stability capacitor to prevent internal LDO oscillation under changing digital loads.

4.3 Sensor Decoupling Capacitors

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- **C7 (0.1 μF SMD):** High-frequency local decoupling capacitor placed directly across the VDD and GND pins of the MAX31865 (U3) to shunt noise and ensure precision analog measurements.
- **C8 (0.1 μF SMD):** High-frequency local decoupling capacitor placed directly across the VDD and VSS pins of the SHT4x (U4).

5 Mechatronic Actuation

Stage-top mechanical actuation is driven by five PWM-controlled servos.

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- **Headers:** Five 1×3 headers (J1–J5) supplying 5V, GND, and a logic signal.
- **Logic Control:** Driven by Pico W pins GPIO0 through GPIO4.
- **Protection:** Each PWM signal line is in series with a $220\,\Omega$ current-limiting resistor (R3–R7) to protect the microcontroller from inductive kickback or stall-current faults.